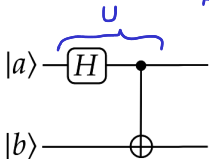


Bell basis

$$a, b \in \{0, 1\}^2 = \mathbb{Z}_2$$



$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle$$

$$X|+\rangle = |-\rangle$$

$$X|-\rangle = |+\rangle$$

$$H|a\rangle = \frac{1}{\sqrt{2}} \sum_{c=0}^1 (-1)^{a \cdot c} |c\rangle$$

$$a_i \cdot c \in \{0, 1\}$$

Computational Basis

a	c	a · c	$(-1)^{a \cdot c}$
0	0	0	+1
0	1	0	+1
1	0	0	+1
1	1	1	-1

$$\begin{aligned} \text{---} \boxed{H} \text{---} & \quad a = (a_1, a_2) \rightarrow |a\rangle = |a_1\rangle |a_2\rangle \\ \text{---} \boxed{H} \text{---} & \quad c = (c_1, c_2) \rightarrow |c\rangle = |c_1\rangle |c_2\rangle \\ H^{\otimes 2} & \quad a \cdot c = a_1 \cdot c_1 \oplus a_2 \cdot c_2 \text{ modulo-2} \end{aligned}$$

Bell basis

$$II|00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |B_1\rangle$$

$$IX|00\rangle = |01\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}} (|01\rangle + |11\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) = |B_2\rangle$$

$$XI|00\rangle = |10\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}} (|00\rangle - |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) = |B_3\rangle$$

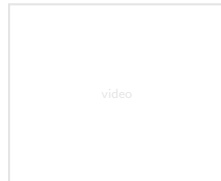
$$XX|00\rangle = |11\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}} (|01\rangle - |11\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) = |B_4\rangle$$

$$P_{red}^2 = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/4 \end{bmatrix}$$

$$\begin{aligned} CNOT(H \otimes I)|a\rangle|b\rangle &= \frac{1}{\sqrt{2}} \sum_c (-1)^{a \cdot c} |c\rangle |b\rangle \\ &= \frac{1}{\sqrt{2}} \sum_c (-1)^{a \cdot c} |c\rangle |b \oplus c\rangle \end{aligned}$$

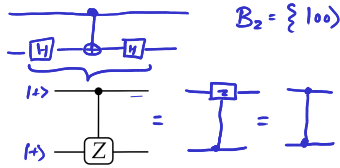
$$\begin{aligned} P &= |B\rangle\langle B| \\ P_{red} &= \text{tr}_{sub}(P) \\ &= \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} = \frac{I}{2} \\ E &\in \cdot \\ S(P_{red}) &= -\left[\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2} \ln \frac{1}{2}\right] \\ &= \ln 2 \\ C &= \sqrt{2(1 - \text{tr}(P_{red}^2))} \\ C &= \sqrt{2(1 - 1/2)} = 1 \end{aligned}$$

Uses: Quantum cryptography, Quantum teleportation, and Superdense coding.

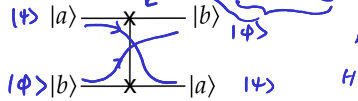


Controlled-Z, SWAP, iSWAP gates

$Z|0\rangle = |0\rangle$
 $Z|1\rangle = -|1\rangle$



$B_2 = \{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$



$i = \sqrt{-1}$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$H = \frac{X + Y}{2} = \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Dici} \begin{bmatrix} 0 & +i \\ +i & 0 \end{bmatrix}$$

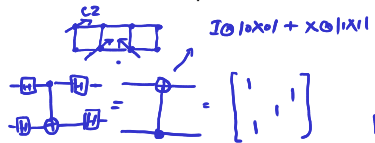
$H^2 \neq I$

$$CZ = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z$$

$$\equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$CZ = I \otimes I - 2|11\rangle\langle 11|$$

Uses: Grover algorithm and Measurement bases quantum computation



$$SWAP = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|$$

$$\equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$iSWAP = \exp(i \frac{\pi}{4} (X \otimes X + Y \otimes Y))$$

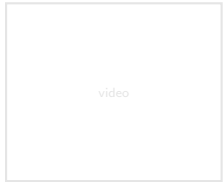
$$\equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Product state-vector into a product state-vector

$$|1\rangle \otimes |\phi\rangle = (\alpha_0|0\rangle + \alpha_1|1\rangle) \otimes (\beta_0|0\rangle + \beta_1|1\rangle)$$

$$= \alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle$$

$$|\phi\rangle \otimes |\psi\rangle = \beta_0\alpha_0|00\rangle + \beta_0\alpha_1|01\rangle + \beta_1\alpha_0|10\rangle + \beta_1\alpha_1|11\rangle$$





Ising coupling gates

$$IJ = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad XX = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad ZZ = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$$

$$R_{xx}(t) = \exp(-it \overbrace{X \otimes X}^H) \quad \begin{matrix} H' = -H \\ = -XX \end{matrix} \Rightarrow$$

$$= (\cos t) I \otimes I - i(\sin t) X \otimes X \quad \Leftarrow$$

$$\equiv \begin{pmatrix} \cos t & 0 & 0 & -i \sin t \\ 0 & \cos t & -i \sin t & 0 \\ 0 & -i \sin t & \cos t & 0 \\ -i \sin t & 0 & 0 & \cos t \end{pmatrix}$$

$$R_{zz}(t) = \exp(-it \overbrace{Z \otimes Z}^H) \quad \begin{matrix} H' = -H \\ = -ZZ \end{matrix} \Rightarrow$$

$$= (\cos t) I \otimes I - i(\sin t) Z \otimes Z \quad \begin{matrix} \uparrow \uparrow \\ \downarrow \downarrow \end{matrix}$$

$$\equiv \begin{pmatrix} e^{-it} & 0 & 0 & 0 \\ 0 & e^{it} & 0 & 0 \\ 0 & 0 & e^{it} & 0 \\ 0 & 0 & 0 & e^{-it} \end{pmatrix}$$

Uses: Measurement bases and trapped-ion quantum computing, Quantum Approximate Optimization Algorithm (QAOA), Variational Quantum Eigensolver (VQE).

$$\underline{H^2 = I}$$

$$\exp(itH) = I + (itH) + \frac{(itH)^2}{2!} + \frac{(itH)^3}{3!} + \frac{(itH)^4}{4!} + \dots \cos(t)$$

$$= I + itH - \frac{t^2}{2!} \underbrace{H^2}_I - i \frac{t^3}{3!} \underbrace{H^3}_H + \frac{t^4}{4!} \underbrace{H^4}_I + \dots = \left[1 - \frac{t^2}{2!} + \frac{t^4}{4!} + \dots \right] I$$

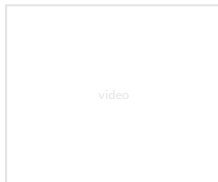
$$+ i \left[t - \frac{t^3}{3!} + \dots \right] H$$

$$\underline{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



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